



SIMATS
ENGINEERING



SIMATS
Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

CSA1017 – SOFTWARE ENGINEERING

INDIVIDUAL REFLECTION REPORT

INTELLIGENT DISASTER EARLY WARNING AND EVACUATION MANAGEMENT SYSTEM

NAME: GANDLURI VEDA VIKAS GOWD

REGISTER NO.: 192472123

ROLE: Testing, Validation, Quality Assurance, and Evidence

DESIGN / DEVELOPMENT DECISION: I prepared functional test cases, invalid-input checks, empty states, map-fallback checks, responsive checks, role-separation checks, and screenshot evidence requirements. I decided to include both successful and failure-oriented scenarios because emergency software must remain understandable when data is missing, sensors are offline, routes are unavailable, or forms are incomplete.

CHALLENGE FACED AND SOLUTION: The most important challenge was maintaining honest evidence boundaries. The prototype uses simulated information, so it would be inappropriate to claim real prediction accuracy, warning latency, or casualty reduction. I addressed this by separating observed, calculated, simulated, and unavailable results in the validation framework.

LEARNING GAINED: I learned that software testing includes more than checking whether buttons work. It also involves traceability, accessibility, safety communication, error recovery, role control, evidence quality, and clear reporting of limitations. This improved my understanding of quality assurance in safety-related applications.

SDG RELEVANCE: The project supports SDG 11 by encouraging dependable and inclusive community-safety tools, while recognizing that actual social impact requires operational evaluation.

COURSE OUTCOME CONTRIBUTION: My work supports the course outcome related to software testing and quality assurance through the test-case design, validation framework, screenshot index, and evidence classification.

POSSIBLE IMPROVEMENT: I would add automated end-to-end testing, performance measurement, security testing, accessibility audits, and structured user acceptance testing.